

Supplementary Material

Excitation-Coherent Telemetry Analysis for Predictive OBD-II Connector Wear Detection: A Physics-Informed Study on Real In-Vehicle Network Data

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This document complements the article. Every number below is generated by `python -m ocwd.experiments.supplement` from the result files produced by the released code.

Contents

S1 Datasets and cleaning	1
S2 Excitation decoders	1
S3 Feature definitions	3
S4 Full detection results	3
S5 Per-vehicle results	3
S6 Robustness to mis-specified physics	3
S7 Prognosis and multi-trip pooling	8
S8 Cross-model generalization	8
S9 Hyperparameter sensitivity	10
S10Transformer baseline	10
S11Statistical significance	10
S12Operating points and calibration	10
S13Window length	10
S14Observability law	10
S15Reproducibility	10

S1 Datasets and cleaning

All healthy telemetry is real and unmodified. Sessions are contiguous recordings after splitting at backward time steps or gaps > 5 s and discarding sessions shorter than 120 s. CANmodes timestamps are repaired for an unpadded millisecond field, a seconds field that lags the millisecond counter after each wrap, and locale thousands separators; rows with non-hexadecimal identifiers are dropped. The four HCRL attack captures share one 482-s benign block, which is kept once; attack segments are excluded. From can-train-and-test only the attack-free drives are used. VED trips shorter than 150 s or with fewer than 20 engine- and vehicle-speed samples are excluded, and at most 40 trips per vehicle are used in the detection corpus.

Table S1: Sessions per vehicle after cleaning.

Vehicle	Corpus	Sessions	Hours	Frames/responses	IDs
Ford Fiesta	sampling	18	3.35	781 310	39
GM Cruze	sampling	34	5.23	1 261 129	79
GM Impala	full-bus	4	1.03	5 362 001	51
GM Silverado	full-bus	4	0.49	4 568 357	98
GM Traverse	full-bus	4	0.33	2 407 121	62
KIA Soul	full-bus	2	0.27	1 879 939	27
MIRGU car	full-bus	1	0.05	368 296	56
Subaru Forester	full-bus	4	0.50	2 873 382	54
VW Gol	sampling	11	2.07	444 502	40
Ford Fiesta	polling	11	2.71	57 613	—
GM Cruze	polling	27	3.61	72 908	—
VW Gol	polling	24	2.71	23 664	—

S2 Excitation decoders

Engine and vehicle speed are decoded only from frames or responses that reached the tool. Values outside 0–8000 rpm and 0–250 km/h are discarded. Polling sessions use PIDs 0x0C and 0x0D.

Table S2: Engine-speed and vehicle-speed decoders (identifier, first byte, byte order, scale).

Vehicle	Engine speed (rpm)	Vehicle speed (km/h)
GM Cruze	0x0C9, byte 1, be, $\times 0.25$	0x3E9, byte 0, be, $\times 0.01609$
VW Gol	0x280, byte 2, le, $\times 0.25$	0x1A0, byte 2, le, $\times 0.005$
KIA Soul	0x316, byte 2, le, $\times 0.25$	0x316, byte 6, u8, $\times 1$
GM Impala	0x0C9, byte 1, be, $\times 0.25$	0x3E9, byte 0, be, $\times 0.01609$
GM Traverse	0x0C9, byte 1, be, $\times 0.25$	0x3E9, byte 0, be, $\times 0.01609$
GM Silverado	0x0C9, byte 1, be, $\times 0.25$	0x3E9, byte 0, be, $\times 0.01609$
Subaru Forester	0x140, byte 2, le14, $\times 1$	0x0D1, byte 0, le, $\times 0.05625$
Ford Fiesta	0x201, byte 0, be, $\times 1$	0x201, byte 4, be, $\times 0.01$

S3 Feature definitions

P denotes the median inter-arrival interval of an identifier (passive) or of the polling loop.

Table S3: ECTA features (25 passive, 20 polling).

Tool	Family	Feature	Definition
passive	loss	rate	frames received per second
passive	loss	n_ids	distinct identifiers in the window
passive	loss	miss_ratio	missing frames / expected frames over periodic identifiers
passive	loss	gap_events_pm	gaps containing ≥ 1 missing frame, per minute
passive	loss	miss_pm	missing frames per minute
passive	loss	max_silence	longest all-identifier silence (s)
passive	loss	silences_pm	all-identifier silences per minute
passive	structure	late_ratio	share of intervals in $(1.08P, 1.5P)$ without a missing frame (retransmission)
passive	structure	bunch_ratio	share of intervals below $0.5P$ (backlog release)
passive	structure	jitter_mad	median $ \Delta/P - 1 $ of regular intervals
passive	structure	sel_chi2	$\log(1 + \chi^2/\text{dof})$ of per-ID losses vs. traffic-proportional losses
passive	structure	sel_gini	Gini coefficient of per-ID loss rates
passive	structure	sel_frac_ids	share of identifiers with any loss
passive	structure	load_at_gap	log ratio of local frame density at losses to window mean
passive	structure	load_corr	Spearman correlation of per-second losses and load
passive	coherence	coh_rho	Spearman $\rho(\text{loss events}, \hat{v})$
passive	coherence	coh_beta	Poisson GLM slope $\hat{\beta}$
passive	coherence	coh_z	Wald statistic z_{β}
passive	coherence	coh_hilo	log rate ratio above/below median excitation
passive	coherence	coh2_rho	as coh_rho, strong events only
passive	coherence	coh2_beta	as coh_beta, strong events only
passive	coherence	coh2_z	as coh_z, strong events only
passive	coherence	coh2_hilo	as coh_hilo, strong events only
passive	coherence	exc_mean	mean excitation proxy $\hat{v}$
passive	coherence	exc_avail	1 if an excitation decoder exists
polling	loss	txn_rate	responses per second
polling	loss	med_dt	median inter-response interval (s)
polling	loss	gap_ratio	share of intervals above median + 0.12 s (timeout-like)
polling	loss	gaps_pm	timeout-like gaps per minute
polling	loss	excess_frac	excess time in timeout-like gaps / window
polling	loss	long_gaps_pm	gaps > 0.8 s per minute (re-initialisation)
polling	loss	max_gap	longest inter-response gap (s)
polling	loss	pid_miss	missing PID cycles / expected cycles
polling	structure	jitter_mad	median $ \Delta/P - 1 $ of regular intervals
polling	structure	gap_dur_med	median excess duration of timeout-like gaps (s)
polling	structure	gap_dur_cv	coefficient of variation of those durations
polling	coherence	coh_rho	Spearman $\rho(\text{loss events}, \hat{v})$
polling	coherence	coh_beta	Poisson GLM slope $\hat{\beta}$
polling	coherence	coh_z	Wald statistic z_{β}
polling	coherence	coh_hilo	log rate ratio above/below median excitation
polling	coherence	coh2_rho	as coh_rho, strong events only
polling	coherence	coh2_beta	as coh_beta, strong events only
polling	coherence	coh2_z	as coh_z, strong events only
polling	coherence	coh2_hilo	as coh_hilo, strong events only
polling	coherence	exc_mean	mean excitation proxy $\hat{v}$

S4 Full detection results

Table S4: Full-bus passive: all methods (6468 windows, 6 vehicles, 6 folds). All = wear vs. all; Inc./Mod./Sev. vs. healthy; ECU/EMI/Ovf. = moderate+severe wear vs. that confounder.

Method	AUROC	AUPRC	TPR@1%	Inc.	Mod.	Sev.	ECU	EMI	Ovf.
ECTA (proposed)	0.880	0.880	0.554	0.555	0.891	0.995	0.949	0.979	0.977
GBM w/o coherence	0.878	0.877	0.560	0.549	0.881	0.996	0.945	0.978	0.977
GBM loss only	0.768	0.763	0.308	0.540	0.761	0.955	0.865	0.935	0.778
GBM loss+coherence	0.812	0.799	0.315	—	—	—	—	—	—
Isolation Forest	0.622	0.569	0.044	0.584	0.939	0.992	0.894	0.762	0.460
One-class SVM	0.596	0.528	0.012	0.544	0.830	0.958	0.828	0.672	0.492
Loss-rate threshold	0.653	0.651	0.280	0.592	0.963	0.996	0.914	0.830	0.546
U-code timeout rule	0.587	0.620	0.285	0.500	0.539	0.922	0.728	0.529	0.718
Coherence test (no training)	0.594	0.589	0.161	0.542	0.684	0.692	0.659	0.638	0.597
1D-CNN (supervised)	0.646	0.669	0.283	0.500	0.522	0.895	0.733	0.712	0.704
LSTM autoencoder	0.631	0.665	0.291	0.500	0.524	0.907	0.715	0.677	0.715

Table S5: Sampling passive: all methods (12684 windows, 3 vehicles, 3 folds). All = wear vs. all; Inc./Mod./Sev. vs. healthy; ECU/EMI/Ovf. = moderate+severe wear vs. that confounder.

Method	AUROC	AUPRC	TPR@1%	Inc.	Mod.	Sev.	ECU	EMI	Ovf.
ECTA (proposed)	0.675	0.702	0.326	0.501	0.556	0.957	0.759	0.766	0.756
GBM w/o coherence	0.674	0.701	0.316	0.500	0.554	0.955	0.758	0.764	0.755
GBM loss only	0.673	0.697	0.287	0.500	0.551	0.952	0.755	0.764	0.752
GBM loss+coherence	0.675	0.701	0.318	—	—	—	—	—	—
Isolation Forest	0.637	0.614	0.110	0.500	0.530	0.883	0.706	0.704	0.707
One-class SVM	0.658	0.661	0.170	0.500	0.542	0.931	0.737	0.737	0.737
Loss-rate threshold	0.667	0.619	0.340	0.500	0.551	0.948	0.750	0.750	0.750
U-code timeout rule	0.668	0.667	0.124	0.501	0.587	0.952	0.763	0.741	0.770
Coherence test (no training)	0.510	0.433	0.010	0.500	0.499	0.532	0.516	0.515	0.516
1D-CNN (supervised)	0.677	0.700	0.345	0.500	0.569	0.964	0.764	0.766	0.767
LSTM autoencoder	0.651	0.625	0.018	0.500	0.534	0.920	0.726	0.727	0.727

Table S6: OBD-II polling: all methods (8631 windows, 3 vehicles, 3 folds). All = wear vs. all; Inc./Mod./Sev. vs. healthy; ECU/EMI/Ovf. = moderate+severe wear vs. that confounder.

Method	AUROC	AUPRC	TPR@1%	Inc.	Mod.	Sev.	ECU	EMI	Ovf.
ECTA (proposed)	0.723	0.727	0.297	0.500	0.528	0.891	0.753	0.878	0.762
GBM w/o coherence	0.686	0.705	0.277	0.500	0.524	0.898	0.761	0.741	0.772
GBM loss only	0.670	0.682	0.250	0.500	0.527	0.845	0.740	0.719	0.753
GBM loss+coherence	0.712	0.708	0.270	—	—	—	—	—	—
Isolation Forest	0.599	0.549	0.068	0.500	0.523	0.821	0.628	0.670	0.656
One-class SVM	0.603	0.564	0.058	0.500	0.511	0.864	0.665	0.618	0.690
Loss-rate threshold	0.579	0.583	0.141	0.500	0.520	0.893	0.598	0.703	0.607
U-code timeout rule	0.609	0.646	0.209	0.500	0.575	0.962	0.690	0.701	0.699
Coherence test (no training)	0.485	0.455	0.026	0.500	0.498	0.413	0.479	0.478	0.461
1D-CNN (supervised)	0.673	0.692	0.272	0.501	0.512	0.882	0.732	0.736	0.750
LSTM autoencoder	0.548	0.503	0.061	0.500	0.551	0.795	0.596	0.611	0.596

Table S7: VED fleet: all methods (236817 windows, 381 vehicles, 5 folds). All = wear vs. all; Inc./Mod./Sev. vs. healthy; ECU/EMI/Ovf. = moderate+severe wear vs. that confounder.

Method	AUROC	AUPRC	TPR@1%	Inc.	Mod.	Sev.	ECU	EMI	Ovf.
ECTA (proposed)	0.775	0.737	0.235	0.500	0.493	0.851	0.750	0.947	0.856
GBM w/o coherence	0.704	0.678	0.190	0.500	0.497	0.845	0.749	0.737	0.825
GBM loss only	0.672	0.643	0.148	0.500	0.501	0.806	0.714	0.708	0.771
GBM loss+coherence	0.757	0.715	0.197	—	—	—	—	—	—
Isolation Forest	0.551	0.496	0.020	0.500	0.494	0.739	0.595	0.570	0.584
One-class SVM	0.541	0.489	0.029	0.500	0.488	0.682	0.592	0.504	0.598
Loss-rate threshold	0.567	0.542	0.073	0.500	0.508	0.813	0.617	0.657	0.560
U-code timeout rule	0.557	0.467	0.006	0.500	0.535	0.796	0.615	0.654	0.533
Coherence test (no training)	0.509	0.442	0.011	0.500	0.496	0.440	0.492	0.606	0.423
1D-CNN (supervised)	0.701	0.698	0.249	0.500	0.539	0.911	0.756	0.734	0.841
LSTM autoencoder	0.554	0.480	0.010	0.500	0.502	0.703	0.592	0.604	0.556

Table S8: ECTA per held-out vehicle (leave-one-vehicle-out).

Corpus	Held-out vehicle	Windows	All	Moderate	Severe
Full-bus passive	GM Impala	2541	0.882	0.900	0.999
Full-bus passive	GM Silverado	1176	0.886	0.863	0.995
Full-bus passive	GM Traverse	798	0.921	0.951	0.982
Full-bus passive	KIA Soul	651	0.869	0.946	0.978
Full-bus passive	MIRGU car	126	0.904	0.898	1.000
Full-bus passive	Subaru Forester	1176	0.910	0.982	0.994
Sampling passive	Ford Fiesta	3990	0.681	0.566	0.954
Sampling passive	GM Cruze	6216	0.684	0.567	0.963
Sampling passive	VW Gol	2478	0.672	0.533	0.961
OBD-II polling	Ford Fiesta	2898	0.786	0.559	0.932
OBD-II polling	GM Cruze	3360	0.782	0.546	0.957
OBD-II polling	VW Gol	2373	0.673	0.529	0.929

S5 Per-vehicle results

S6 Robustness to mis-specified physics

Trained on the nominal model, tested on held-out vehicles generated with each perturbation.

Table S9: Robustness, Full-bus passive (AUROC vs. healthy and confounders).

Test physics	All	Incipient	Moderate	Severe
nominal	0.901	0.766	0.940	0.997
sigma=0.5	0.851	0.701	0.858	0.995
sigma=1.2	0.927	0.833	0.956	0.994
d0 x0.25	0.864	0.711	0.895	0.986
d0 x4	0.916	0.788	0.966	0.995
nu0 x0.5	0.875	0.717	0.909	0.998
nu0 x2	0.905	0.771	0.951	0.994
R_th=3	0.948	0.862	0.989	0.995
R_th=15	0.837	0.693	0.837	0.982
p_power=0.1	0.901	0.772	0.937	0.994
p_power=0.6	0.886	0.752	0.916	0.992
holdup=0.5ms	0.906	0.766	0.957	0.996
holdup=10ms	0.901	0.766	0.940	0.996
road noise 0	0.888	0.741	0.926	0.996
road noise x2	0.893	0.749	0.932	0.999
engine-only excitation	0.893	0.737	0.941	1.000
road-only excitation	0.875	0.739	0.918	0.968

S7 Prognosis and multi-trip pooling

S8 Cross-model generalization

The test faults are generated by a structurally different process: a two-state (Gilbert–Elliott) contact alternating between quiet periods and chatter episodes (mean 0.1 s, 100 interruptions/s), with exponentially distributed durations and a brown-out triggered when the *cumulative* open time of a supply-pin episode exceeds the hold-up time. Expected interruption rate and mean duration equal those of the nominal Poisson model; temporal clustering, duration law and brown-out mechanism differ. The detector is trained only on the nominal model.

Table S10: Robustness, VED fleet (AUROC vs. healthy and confounders).

Test physics	All	Incipient	Moderate	Severe
nominal	0.748	0.686	0.670	0.887
sigma=0.5	0.754	0.688	0.679	0.895
sigma=1.2	0.745	0.685	0.667	0.882
d0 x0.25	0.660	0.667	0.663	0.649
d0 x4	0.774	0.661	0.728	0.934
nu0 x0.5	0.731	0.682	0.672	0.838
nu0 x2	0.761	0.688	0.672	0.922
R_th=3	0.792	0.677	0.731	0.966
R_th=15	0.688	0.671	0.670	0.722
p_power=0.1	0.713	0.682	0.670	0.788
p_power=0.6	0.759	0.681	0.663	0.934
holdup=0.5ms	0.774	0.660	0.728	0.935
holdup=10ms	0.653	0.670	0.666	0.623
road noise 0	0.754	0.694	0.680	0.888
road noise x2	0.746	0.684	0.666	0.887
engine-only excitation	0.745	0.690	0.675	0.870
road-only excitation	0.746	0.684	0.669	0.885

Table S11: VED vehicle-level detection after pooling K trips (left: vs. all; right: vs. healthy).

K	Vehicles	Inc.	Mod.	Sev.	Inc.	Mod.	Sev.
1	381	0.718	0.704	0.907	0.500	0.483	0.850
2	368	0.759	0.753	0.965	0.487	0.478	0.934
3	354	0.782	0.767	0.987	0.484	0.461	0.971
5	341	0.809	0.802	0.995	0.502	0.489	0.989
10	320	0.830	0.817	1.000	0.500	0.470	0.999
20	272	0.844	0.835	1.000	0.503	0.483	1.000

Table S12: Prognosis, failure at 6 Ω (110 of 166 vehicles fail; health-index $R^2 = 0.86$).

Method	MAE (d)	Median AE (d)	$\alpha=0.3$ acc. %	Timely %	Prem. %	Missed %	Lead (d)	FA %
proposed	71	51	24	13	29	58	0.0	16.1
raw_hi	72	54	25	5	7	87	0.0	1.8
conditional	64	44	26	7	7	85	0.0	0.0
fleet	66	44	19	36	28	35	18.7	60.7
linear	–	–	2	8	10	82	0.0	3.6

Table S13: Prognosis, failure at 20 Ω (100 of 166 vehicles fail; health-index $R^2 = 0.86$).

Method	MAE (d)	Median AE (d)	$\alpha=0.3$ acc. %	Timely %	Prem. %	Missed %	Lead (d)	FA %
proposed	71	51	24	45	28	27	4.0	12.1
raw_hi	79	61	26	2	6	92	0.0	1.5
conditional	67	46	26	5	10	85	0.0	0.0
fleet	69	47	18	36	34	30	25.1	60.6
linear	–	–	3	88	5	7	6.4	1.5

Table S14: Train on the nominal model, test on the nominal or the Markov model (vehicle-disjoint).

Corpus	Train $\rightarrow$ test	All	Incipient	Moderate	Severe
Full-bus passive	nominal $\rightarrow$ nominal	0.880	0.555	0.891	0.995
Full-bus passive	nominal $\rightarrow$ Markov	0.856	0.528	0.848	0.999
OBD-II polling	nominal $\rightarrow$ nominal	0.723	0.500	0.528	0.891
OBD-II polling	nominal $\rightarrow$ Markov	0.779	0.501	0.675	0.941
VED fleet	nominal $\rightarrow$ nominal	0.771	0.500	0.490	0.842
VED fleet	nominal $\rightarrow$ Markov	0.813	0.499	0.577	0.950

S9 Hyperparameter sensitivity

Table S15: ECTA with different gradient-boosting settings (same folds).

Corpus	Setting	All	Moderate	Severe
Full-bus passive	default (400 trees, 31 leaves, lr 0.05)	0.880	0.891	0.995
Full-bus passive	200 trees	0.881	0.891	0.995
Full-bus passive	800 trees	0.880	0.895	0.994
Full-bus passive	15 leaves	0.877	0.880	0.995
Full-bus passive	63 leaves	0.877	0.885	0.995
Full-bus passive	lr 0.1	0.880	0.887	0.995
Full-bus passive	lr 0.02, 1000 trees	0.881	0.890	0.996
OBD-II polling	default (400 trees, 31 leaves, lr 0.05)	0.723	0.528	0.891
OBD-II polling	200 trees	0.720	0.526	0.888
OBD-II polling	800 trees	0.727	0.530	0.894
OBD-II polling	15 leaves	0.721	0.528	0.890
OBD-II polling	63 leaves	0.723	0.527	0.893
OBD-II polling	lr 0.1	0.722	0.529	0.892
OBD-II polling	lr 0.02, 1000 trees	0.723	0.527	0.890
VED fleet	default (400 trees, 31 leaves, lr 0.05)	0.771	0.492	0.842
VED fleet	200 trees	0.762	0.489	0.835
VED fleet	800 trees	0.775	0.493	0.846
VED fleet	15 leaves	0.760	0.486	0.833
VED fleet	63 leaves	0.775	0.494	0.846
VED fleet	lr 0.1	0.773	0.493	0.846
VED fleet	lr 0.02, 1000 trees	0.771	0.490	0.842

S10 Transformer baseline

A 2-layer, 4-head Transformer encoder ($d = 32$, temporal average pooling to ≤ 60 tokens) is trained end-to-end on the binned sequences with the same folds as all other methods. Because of its CPU cost it is trained on at most 10 000 windows per fold for 8 epochs (the 1D-CNN: 40 000 windows, 15 epochs).

Table S16: Transformer baseline vs. ECTA and the 1D-CNN (AUROC).

Corpus	Transformer	95% CI	Moderate	Severe	ECTA	1D-CNN
Full-bus passive	0.641	[0.634, 0.645]	0.528	0.907	0.880	0.646
Sampling passive	0.665	[0.660, 0.676]	0.555	0.942	0.675	0.677
OBD-II polling	0.665	[0.599, 0.748]	0.514	0.862	0.723	0.673
VED fleet	0.590	[0.588, 0.592]	0.499	0.757	0.775	0.701

S11 Statistical significance

Paired bootstrap over vehicles (1 000 resamples) of $\Delta\text{AUROC} = \text{ECTA} - \text{baseline (wear vs. all)}$; two-sided p-values, whose resolution is 0.001. The sampling and polling corpora contain only three vehicles, so their vehicle-level bootstrap is coarse and their p-values should be read as indicative; the full-bus (six vehicles) and VED (381 vehicles) tests are the informative ones.

S12 Operating points and calibration

The threshold is set to reach the stated detection rate for the given wear stage; false alarms are counted on the real healthy windows of held-out vehicles.

S13 Window length

S14 Observability law

S15 Reproducibility

Software: Python 3.11, NumPy 2.4.6, pandas 3.0.6, SciPy 1.17.1, scikit-learn 1.9.1, LightGBM 4.7.0, PyTorch 2.14.1 (CPU). Hardware: 4-core CPU, no GPU. Complete pipeline: `ocwd/get_data.sh` then `ocwd/run_all.sh` (about 4–5 h). Fourteen unit tests (`ocwd/tests`) cover the timestamp repair, interruption statistics, loss mechanics, Poisson coherence test, Markov fault model and baselines. Random seeds: three injection seeds (0, 1, 2) for all detection corpora; per-session streams are derived from CRC32 of the session name, so results are independent of execution order.

Table S17: ECTA vs. every baseline.

Corpus	Baseline	Δ AUROC	95% CI	p
Full-bus passive	GBM w/o coherence	+0.002	[-0.001, +0.004]	0.298
Full-bus passive	GBM loss only	+0.112	[+0.070, +0.136]	≤ 0.001
Full-bus passive	GBM loss+coherence	+0.068	[+0.049, +0.088]	≤ 0.001
Full-bus passive	Isolation Forest	+0.258	[+0.255, +0.281]	≤ 0.001
Full-bus passive	One-class SVM	+0.284	[+0.258, +0.298]	≤ 0.001
Full-bus passive	Loss-rate threshold	+0.227	[+0.224, +0.250]	≤ 0.001
Full-bus passive	U-code timeout rule	+0.293	[+0.290, +0.314]	≤ 0.001
Full-bus passive	Coherence test (no training)	+0.287	[+0.278, +0.302]	≤ 0.001
Full-bus passive	1D-CNN (supervised)	+0.234	[+0.223, +0.260]	≤ 0.001
Full-bus passive	LSTM autoencoder	+0.249	[+0.237, +0.277]	≤ 0.001
Full-bus passive	Transformer (supervised)	+0.239	[+0.235, +0.266]	≤ 0.001
Sampling passive	GBM w/o coherence	+0.001	[-0.003, +0.004]	0.494
Sampling passive	GBM loss only	+0.002	[-0.006, +0.005]	0.494
Sampling passive	GBM loss+coherence	+0.000	[-0.002, +0.001]	0.494
Sampling passive	Isolation Forest	+0.039	[+0.024, +0.054]	≤ 0.001
Sampling passive	One-class SVM	+0.018	[+0.009, +0.030]	≤ 0.001
Sampling passive	Loss-rate threshold	+0.009	[+0.005, +0.016]	≤ 0.001
Sampling passive	U-code timeout rule	+0.007	[+0.004, +0.014]	≤ 0.001
Sampling passive	Coherence test (no training)	+0.165	[+0.155, +0.179]	≤ 0.001
Sampling passive	1D-CNN (supervised)	-0.002	[-0.025, +0.011]	0.998
Sampling passive	LSTM autoencoder	+0.024	[+0.008, +0.035]	≤ 0.001
Sampling passive	Transformer (supervised)	+0.011	[+0.004, +0.020]	≤ 0.001
OBD-II polling	GBM w/o coherence	+0.037	[+0.020, +0.044]	≤ 0.001
OBD-II polling	GBM loss only	+0.053	[+0.024, +0.060]	≤ 0.001
OBD-II polling	GBM loss+coherence	+0.011	[+0.001, +0.018]	≤ 0.001
OBD-II polling	Isolation Forest	+0.124	[+0.089, +0.145]	≤ 0.001
OBD-II polling	One-class SVM	+0.120	[+0.107, +0.162]	≤ 0.001
OBD-II polling	Loss-rate threshold	+0.144	[+0.098, +0.216]	≤ 0.001
OBD-II polling	U-code timeout rule	+0.114	[+0.056, +0.211]	≤ 0.001
OBD-II polling	Coherence test (no training)	+0.238	[+0.191, +0.300]	≤ 0.001
OBD-II polling	1D-CNN (supervised)	+0.051	[+0.018, +0.051]	≤ 0.001
OBD-II polling	LSTM autoencoder	+0.175	[+0.093, +0.212]	≤ 0.001
OBD-II polling	Transformer (supervised)	+0.058	[+0.030, +0.074]	≤ 0.001
VED fleet	GBM w/o coherence	+0.070	[+0.068, +0.073]	≤ 0.001
VED fleet	GBM loss only	+0.102	[+0.100, +0.105]	≤ 0.001
VED fleet	GBM loss+coherence	+0.018	[+0.017, +0.019]	≤ 0.001
VED fleet	Isolation Forest	+0.224	[+0.221, +0.227]	≤ 0.001
VED fleet	One-class SVM	+0.233	[+0.230, +0.236]	≤ 0.001
VED fleet	Loss-rate threshold	+0.208	[+0.205, +0.211]	≤ 0.001
VED fleet	U-code timeout rule	+0.218	[+0.214, +0.222]	≤ 0.001
VED fleet	Coherence test (no training)	+0.266	[+0.261, +0.272]	≤ 0.001
VED fleet	1D-CNN (supervised)	+0.073	[+0.071, +0.076]	≤ 0.001
VED fleet	LSTM autoencoder	+0.221	[+0.218, +0.224]	≤ 0.001
VED fleet	Transformer (supervised)	+0.185	[+0.181, +0.189]	≤ 0.001

Table S18: False alarms at fixed detection rates.

Corpus	Stage	Detection	FPR healthy	False alarms / h	FPR confounders
Full-bus passive	moderate	80%	11.0%	13.25	6.6%
Full-bus passive	moderate	90%	52.6%	63.12	25.3%
Full-bus passive	moderate	95%	66.6%	79.87	34.1%
Full-bus passive	severe	80%	0.0%	0.00	0.0%
Full-bus passive	severe	90%	0.0%	0.00	0.1%
Full-bus passive	severe	95%	0.3%	0.39	0.4%
Sampling passive	moderate	80%	76.5%	45.89	75.6%
Sampling passive	moderate	90%	88.1%	52.85	86.6%
Sampling passive	moderate	95%	94.2%	56.52	92.6%
Sampling passive	severe	80%	0.3%	0.20	0.3%
Sampling passive	severe	90%	2.6%	1.59	2.6%
Sampling passive	severe	95%	44.2%	26.52	43.0%
OBD-II polling	moderate	80%	79.3%	23.80	65.3%
OBD-II polling	moderate	90%	89.5%	26.86	77.4%
OBD-II polling	moderate	95%	94.6%	28.39	83.7%
OBD-II polling	severe	80%	1.2%	0.36	1.0%
OBD-II polling	severe	90%	66.2%	19.85	41.8%
OBD-II polling	severe	95%	74.2%	22.26	60.1%
VED fleet	moderate	80%	83.3%	7.10	35.7%
VED fleet	moderate	90%	92.3%	7.86	45.1%
VED fleet	moderate	95%	96.2%	8.20	52.8%
VED fleet	severe	80%	19.7%	1.68	8.4%
VED fleet	severe	90%	78.9%	6.73	32.7%
VED fleet	severe	95%	93.2%	7.94	46.7%

Table S19: Calibration of the ECTA wear probability (wear vs. all).

Corpus	Brier	ECE
Full-bus passive	0.145	0.098
Sampling passive	0.312	0.276
OBD-II polling	0.237	0.180
VED fleet	0.185	0.052

Table S20: Full-bus detection vs. window length (one seed, leave-one-vehicle-out).

Window	Windows	All	Incipient	Moderate	Severe
10s	6629	0.824	0.543	0.843	0.983
30s	2156	0.879	0.565	0.895	0.995
60s	1050	0.911	0.643	0.930	0.999

Table S21: Inputs and outputs of the observability analysis (Spearman $\rho = 0.93$).

Corpus	Stage	R (Ω)	Unit rate (1/s)	A	σ_B	SNR	Best AUROC
Full-bus passive	incipient	0.67	1613.5	0.206	49.7	0.00414	0.592
Full-bus passive	moderate	3.00	1613.5	44.8	49.7	0.9	0.963
Full-bus passive	severe	13.42	1613.5	6.95e+03	49.7	140	0.996
Sampling passive	incipient	0.67	65.6	6.75e-07	0.0909	7.43e-06	0.501
Sampling passive	moderate	3.00	65.6	0.0282	0.0909	0.31	0.587
Sampling passive	severe	13.42	65.6	9.31	0.0909	102	0.964
OBD-II polling	incipient	0.67	5.6	0.0108	33.9	0.000319	0.501
OBD-II polling	moderate	3.00	5.6	1.18	33.9	0.0348	0.575
OBD-II polling	severe	13.42	5.6	29.5	33.9	0.869	0.962
VED fleet	incipient	0.67	1.3	0.00785	54.5	0.000144	0.500
VED fleet	moderate	3.00	1.3	0.992	54.5	0.0182	0.539
VED fleet	severe	13.42	1.3	66.1	54.5	1.21	0.911